

Task-1

Date: 05/08/2025

1.1: Karan spent ₹150 on books, ₹220 on groceries, and ₹90 on transport. Help him calculate the total expenses.

Aim: To write a python program that calculates the total amount spent by karan on books groceries, and transport

Algorithm:

1. Start the program
2. Accept the amount spent on books, groceries, and transport
3. Calculate the total expenses by summing all three amounts
4. Display the total amount spent
5. End the program.

Result: Thus, the amount spent by Karan on books, groceries and transport are proved

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python program :-

program to calculate total expenses of karan.

step 1: Assign expenses

books = 150

groceries = 220

transport = 90

step 2: calculate total

total - expense = books + groceries + transport

step 3: Display the results

print("Total expenses incurred by karan : ₹", total - expense)

Sample Input :-

(values are already assigned in the program - no manual input required)

Books = ₹ 150

Groceries = ₹ 220

Transport = ₹ 90

Sample output :-

Total expenses incurred by karan = ₹ 460

Date: 06/08/2025

1.2 write a BMI calculator. Ask the user for weight(kg) and height (m), then calculate and display their BMI.

Aim : To write a python program that calculates and displays the body mass index (BMI) of a person using their weight (in kilograms) and height (in meters).

Algorithm:

- 1) start the program
- 2) prompt the user to input their weight in kilograms.
- 3) prompt the user to input their height in meters.
- 4) calculate BMI using the formula

$$BMI = \frac{\text{weight}}{\text{height}^2}$$

- 5) Display the calculated BMI
- 6) End the program

Result: Thus, the body mass index of a person using their weight (kg) and height (m) are proved

Python program

BMI calculator

step 1: Get input from the user

weight = float(input("Enter your ~~he~~ weight in kilograms:"))

height = float(input("Enter your height in meters:"))

step 2: calculate BMI

bmi = weight / (height**2)

step 3: Display result

print("your body mass Index (BMI) is:", round(bmi, 2))

Sample input:

Enter your weight in kilograms: 70

Enter your height in meters: 1.75

Sample output:

your Body mass Index (BMI) is: 22.86

Date: 06/05/2025

1.3 Laya wants to calculate the area of a scalene triangle with sides of length 8cm, 6cm, and 4cm, Help her write a python program that computes the area using Heron's formula.

Aim: To write a python program to find the area of a triangle when the lengths of all three sides are given, using, Heron's formula.

Algorithm:

- 1) start the program
- 2) Accept or assign the lengths of the three sides : a, b and c
- 3) Calculate the semi-perimeter.

$$s = \frac{a+b+c}{2}$$

- 4) use Heron's formula to calculate the area:

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

- 5) Display the area of the triangle
- 6) End the program.

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Result: Thus, the area of triangle when the lengths of all three sides are proved by heron's formula.

python program:

```
import math
```

step 1: Assign side lengths

```
a=8
```

```
b=6
```

```
c=4
```

step 2: calculate semi-perimeter

$$s = (a+b+c)/2$$

step 3: Apply Heron's formula

$$\text{area} = \text{math.sqrt}(s * (s-a) * (s-b) * (s-c))$$

~~area~~

step 4: Display result

```
print("The area of the triangle is: ", round(area, 2), "Square cm")
```

Sample Input:

(values are already assigned)

side a = 8cm

side b = 6cm

side c = 4cm

Sample output:

The area of the triangle is : 11.62 square cm.